

Hamid Shafieasl

Ph.D. Student in Computer Science (Artificial Intelligence / Data Science)

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Professional Summary

Ph.D. student in Computer Science specializing in Artificial Intelligence, with a previous Ph.D. in Mathematics focused on Harmonic Analysis. My work lies at the intersection of mathematics and computer science, with particular interests in high-dimensional data analysis, machine learning, computational geometry, graph algorithms, geometric shape analysis, and reinforcement learning. My research combines mathematical structure with algorithmic and computational methods for analyzing complex data and geometric objects.

Research Interests

- High-dimensional data analysis and large-scale computation
- Machine learning and reinforcement learning
- Computational geometry and geometric shape analysis
- Graph algorithms and spectral methods
- Mathematical foundations and mathematical interfaces of machine learning

Education

- **Ph.D. in Computer Science (Artificial Intelligence)**, University of Utah *Expected 2029*
Graduate coursework includes: High-Dimensional Data Analysis, Data Mining, Artificial Intelligence, Machine Learning, and Computational Topology.
- **Ph.D. in Mathematics**, Azerbaijan Shahid Madani University 2019
Advanced coursework included Calculus of Variations, Convex Optimization, Functional Analysis, Operator Algebras, Measure Algebras on Groups and Semigroups, and Amenability of Groups and Semigroups.
- **M.Sc. in Mathematics**, Azerbaijan Shahid Madani University 2011
Coursework included Abstract Harmonic Analysis, Functional Analysis, Real Analysis, Complex Analysis, Advanced Algebra, and Algebraic Topology.
Ranked 1st in cumulative GPA.
- **B.Sc. in Mathematics**, Azad University of Tabriz 2009
Coursework included Linear Algebra, Discrete Mathematics, and Principles of Programming Languages.
Ranked 1st in cumulative GPA.

Technical Skills

Programming: Python, object-oriented programming, numerical and scientific computing.

Scientific Computing: NumPy, SciPy, Matplotlib.

Machine Learning & AI: supervised and unsupervised learning, reinforcement learning, classification, feature extraction, model evaluation, search methods, and algorithmic reasoning.

Data & Graph Algorithms: high-dimensional data analysis, signal sampling, noise reduction, PageRank, Dijkstra's algorithm, graph-based methods, and large-scale computation.

Computational Geometry & Shape Analysis: geometric reasoning, rotation-invariant shape represen-

tation, planar object comparison, star-shaped and non-star-shaped object analysis, discrete curve geometry, curvature, torsion, and geometric similarity methods.

Mathematics: harmonic analysis, linear algebra, probability, real and complex analysis, differential equations, functional analysis, topology, optimization, and discrete mathematics.

Geometric Modeling Tools: Rhino–Grasshopper, Python scripting, and computational geometric modeling.

Research & Projects

- **Rotation-Invariant Shape Representation**

Developed computational geometry and similarity-function methods for rotation-invariant representation and comparison of planar shapes. The work combines mathematical descriptors with computational methods for geometric object comparison and classification.

[Interactive Project](#) | [arXiv Paper](#)

- **Galaxy Shape Geometry**

Developed computational geometry and similarity-based methods for studying and comparing complex geometric structures, including irregular and nontrivial planar shapes.

[Interactive Project](#)

- **Signature Tool**

Developed an interactive computational tool for constructing geometric signatures and analyzing similarity between shapes.

[University-Hosted Interactive Tool](#)

- **Computational Geometry of Star-Shaped and Non-Star-Shaped Objects**

Developed an interdisciplinary framework for analyzing star-shaped and non-star-shaped objects in two-dimensional space. The approach connects mathematical structure with computational geometry applications including shape recognition, boundary representation, and geometric comparison.

- **Geometric Origami Modeling**

Used Python and Rhino–Grasshopper to study and model curved origami structures. The work involved geometric parameterization, algorithmic modeling, and computational analysis and contributed to published research on curved origami design.

- **Discrete Curve Geometry**

Investigated curvature, torsion, and other geometric properties of discretely sampled curves, with applications to computer-aided design, computer graphics, and geometric modeling.

Professional Experience

- **Python Instructor**, Private Institutes 2020–Present
 - Taught Python programming with emphasis on data workflows, algorithm design, numerical computation, and machine learning fundamentals.
- **Mathematics Lecturer**, Azerbaijan Shahid Madani University 2010–2018
 - Taught Linear Algebra, Calculus, Mathematical Analysis, Differential Equations, Complex Analysis, and Discrete Mathematics.
 - Guided students in rigorous mathematical reasoning and its application to computational and algorithmic problems.
- **Mathematics Teacher**, Various Colleges 2009–2024
 - Taught advanced mathematics with emphasis on abstraction, quantitative reasoning, rigor, and analytical problem-solving.

Publications

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1. **H. Shafieasl**, J. M. Phillips. “Rotation-Invariant Vectorized Shape Representations.” *arXiv preprint*, 2026. [\[arXiv\]](#)
 2. **H. Shafieasl**, A. M. Tavakkoli. “On the extension of inner derivations from dense ideals in Banach algebras.” *arXiv preprint*, 2026. [\[arXiv\]](#)
 3. A. Safary, **H. Shafieasl**, J. Mitani. “Parameterized Folded State Shape Modeling of David Huffman’s Ellipse.” *Journal for Geometry and Graphics*, 28(2):199–212, 2025. [\[Link\]](#)
 4. A. Safary, **H. Shafieasl**, J. Mitani. “Geometric Design Tool for One-Fold, a Curved Origami with a Single Fold.” *Journal for Geometry and Graphics*, 28(1):89–101, 2024. [\[Link\]](#)
 5. H. Pourmahmood-Aghababa, M. H. Sattari, **H. Shafieasl**. “Bounded Pseudo-Amenability and Contractibility of Certain Banach Algebras.” *Filomat*, 34(5):1701–1712, 2020. [\[PDF\]](#)
 6. **H. Shafieasl**. “A Note on Ramified Coverings of Riemann Surfaces.” *International Mathematical Forum*, 15(8):377–381, 2020. [\[PDF\]](#)
 7. **H. Shafieasl**, M. H. Sattari. “Symmetric Module and Connes Amenability.” *Sahand Communications in Mathematical Analysis*, 5(1):49–59, 2017. [\[Link\]](#)
 8. M. H. Sattari, **H. Shafieasl**. “Product of Derivations on Triangular Banach Algebras.” *Journal of Hyperstructures*, 6(1), 2017. [\[PDF\]](#)

Awards & Honors

- Ranked 1st in cumulative GPA, B.Sc. in Mathematics.
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